aqualytic
Environmental & water research labs

LABORATORY TEST REPORT: AQL-LMS-A-780-001

KENYA ACCREDITATION SERVICE KENAS ISO/IEC

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LABORATORY TEST REPORT

FEBRUARY 26TH, 2021

Description of Sample: PURIFIED WATER
Sample Source: GARISSA
Submitted By: ASHYAA INVESTMENT LTD
Customer Contact: ABSHIR MOHAMED
Sampled By: CLIENT

Date of Sampling: 23/02/2021
Date sample Received: 24/02/2021
Date of Analysis: 25/02/2021
Date of Report Issue: 26/02/2021
Sample ID: S0513-16201

PARAMETERS	METHOD	VALUES	KS EAS 153:2018	WHO STANDARDS	REMARKS
PHYSICAL TESTS					
pH (pH Units)	AQL/TM/PHY/02	7.9	6.5 – 8.5	6.5 – 8.5	PASS
Color (Pt. Co. APHA)	AQL/TM/HACH/8025	<5	15	15	PASS
Turbidity (NTU)	AQL/TM/HACH/2100Q	0.5	1.0	5.0	PASS
Total Suspended Solids (mg/L)	AQL/TM/HACH/8006	ND	NIL	NIL	PASS
Total Dissolved Solids (mg/L)	AQL/TM/PHY/01	83.1	1000	1500	PASS
Conductivity (µS/cm)	AQL/TM/HACH/8160	170.8	1500	NS	PASS
CHEMICAL TESTS					
P. Alkalinity (mg/L $CaCO_3$)	AQL/TM/HACH/ISO/9963-1:1994	ND	NS	NS	-
Total Alkalinity (mg/L $CaCO_3$)	AQL/TM/HACH/ISO/9963-1:1994	72.6	NS	500	-
Chloride (mg/L Cl^-)	AQL/TM/HACH/ISO/9297:2000	<5	250	250	PASS
Fluoride (mg/L F)✔	AQL/TM/CHEM/02	0.27	1.5	1.5	PASS
Sulfate (mg/L SO_4^{2-})	AQL/TM/HACH/8051	8	400	450	PASS
Nitrate (mg/L NO_3)	AQL/TM/HACH/8039	2.1	45	10	PASS
Phosphate (mg/L PO_4^{-3})✔	AQL/TM/CHEM/01	0.31	2.2	NS	PASS
Sodium (mg/L Na)	AQL/TM/HACH/8322	14.9	200	200	PASS
Calcium (mg/L Ca^{2+})✔	AQL/TM/CHEM/07	0.02	150	100	PASS
Magnesium (mg/L Mg^2) ✔	AQL/TM/CHEM/07	074	100	100	PASS
Iron (mg/L Fe^{2+})	AQL/TM/HACH/8008	<0.002	0.3	0.3	PASS
Manganese (mg/L Mn)	AQL/TM/HACH/8034	0.12	0.1	0.1	PASS
Ammonia (mg/L NH_3)	AQL/TM/HACH/8155	<0.01	0.5	1.5	PASS
Copper (mg/L Cu)✔	AQL/TM/CHEM/06	<0.04	2.0	2.0	PASS
Zinc (mg/L Zn)	AQL/TM/HACH/8009	0.04	5.0	3.0	PASS
OTHER PARAMETERS					
Ca. Hardness (mg/L Ca^{2+} $CaCO_3$) ✔	AQL/TM/CHEM/07	0.05	NS	NS	-
Mg. Hardness (mg/L Mg^{2+} $CaCO_3$) ✔	AQL/TM/CHEM/07	3.04	NS	NS	-
Total Hardness (mg/L $CaCO_3$) ✔	AQL/TM/CHEM/07	3.09	300	500	PASS
Silica (mg/L SiO_2)	AQL/TM/HACH/8185	19.6	NS	NS	-

ISE: Ion Selective Electrode, NS: No Set Standard, ND: Not Detected, KS: Kenya Standard, EAS: East African Standard, Parameters marked with (✓) are accredited while those marked with (*) are subcontracted.

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The results relate to the sample(s) submitted. The laboratory will not be held responsible for any sampling error.

COMMENTS

Based on the tested parameters only, the water sample was found to conform to the packaging drinking water specifications (KS EAS 153:2018).

LABAN MACHARIA
ASSISTANT TECHNICAL MANAGER

JOHN KIIRU
QUALITY MANAGER

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BACTERIOLOGICAL ANALYSIS

TYPE OF MICRO-ORGANISM	UNIT	TEST METHOD	RESULTS TREATED AT 37 °C	KS EAS 153:2018 LIMIT
Total Coliforms	**CFU/100 ml**	**KS ISO 9308-1:2014**	**715**	**Not detectable**
Escherichia coli	CFU/100 ml	KS ISO 9308-1:2014	ND	Not detectable
Total Viable Counts	**CFU/ ml**	**KS ISO 6222:1999**	**TN**	**50**

CFU: Colony Forming Units
KS: Kenya Standard
EAS: East African Standard
ND: Not Detected
TN: Too Numerous to count
Parameter marked with (✓) is accredited and those marked with (*) are subcontracted.

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COMMENTS

Based on the tested parameters only, the water sample was found not to conform to the packaging drinking water specifications (KS EAS 153:2018). The number of colony forming units for total coliforms and total viable counts does not meet the set standard. Therefore, further treatment is recommended.

Okoyaya For
LABAN MACHARIA
ASSISTANT TECHNICAL MANAGER

JOHN KIIRU
QUALITY MANAGER

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